

Cohen-Macaulayness depends on the characteristic of k .

Let $\Delta \approx \mathbb{RP}^n$ $n \geq 2$

$$\dim \tilde{H}_i(\mathbb{RP}^n; k) = \begin{cases} 0 & \text{if } \text{char } k \neq 2 \\ 1 & \text{if } \text{char } k = 2 \end{cases} \text{ for } 0 < i < n$$

Δ is Cohen-Macaulay if $\text{char } k \neq 2$, and not if $\text{char } k = 2$

Call a sequence $h_0, h_1, \dots, h_r$ an **O-sequence** if there is a finite-dimensional graded K -algebra A generated in degree 1 such that

$$H(A, t) = h_0 + h_1 t + \dots + h_r t^r$$

$$h_k \geq 0$$

$$h_0 = 1$$

$$h_k \leq \binom{h_1 + k - 1}{k}$$

For any positive integers n and i , there is a unique way to write

$$n = \binom{n_i}{i} + \binom{n_{i-1}}{i-1} + \dots + \binom{n_j}{j}$$

where $n_i > n_{i-1} > \dots > n_j \geq j \geq 1$

Define

$$n^{<i>} = \binom{n_i+1}{i+1} + \binom{n_{i-1}+1}{i} + \dots + \binom{n_j+1}{j+1}$$

$$0^{<i>} = 0$$

Thm (Macaulay): A finite sequence $h_0, h_1, \dots, h_r$ is an $\mathcal{O}$ -sequence if and only if

$$h_0 = 1$$

$$0 \leq h_{i+1} \leq h_i^{\langle i \rangle} \quad \forall i \geq 1$$

If Δ is simplicial sphere with h -vector $h_0, \dots, h_d$ then

- $h_0, \dots, h_d$ is an O -sequence
- $h_k = h_{d-k}$ Dehn-Sommerville equations

Are all such sequences h -vectors of simplicial spheres?

1, 4, 3, 4, 3, 4, 1 is an O-sequence but not the h-vector of a simplicial sphere.

Lower Bound theorem: If $h_0, \dots, h_d$ is the h-vector of a simplicial sphere, then $h_i \leq h_k$ for all $1 \leq k \leq d-1$.

Generalized lower bound theorem:

$$h_0 \leq h_1 \leq h_2 \leq \dots \leq h_{\lfloor d/2 \rfloor}$$

$$h_{k+1} - h_k$$

1, 3, 3, 4, 3, 3, 1 is an O-sequence but
not the h-vector of a
simplicial sphere.

Let $A = A_0 \oplus A_1 \oplus \dots \oplus A_d$ be a graded K -algebra generated in degree 1.

We say that an element $l \in A_1$ is a **Lefschetz element** if the multiplication map

$$l^{d-2k} : A_k \rightarrow A_{d-k}$$

is an isomorphism for all $k < d/2$.

We say A has the **hard Lefschetz property** if it has a Lefschetz element.

Suppose l is a Lefschetz element for A .

Then the multiplication map $l: A_k \rightarrow A_{k+1}$ is injective for all $k < d/2$, and surjective for all $k \geq d/2$.

Let $B = A/lA$

$$\widetilde{A}_0 \quad A_1 \quad A_2 \quad A_3 \quad A_4 \quad A_5 \quad A_6 \quad lA$$

Then

$$H(LB, t) = h_0 + (h_1 - h_0)t + (h_2 - h_1)t^2 + (h_3 - h_2)t^3 \\ + \dots (h_{Ld/2} - h_{Ld/2-1})t^{Ld/2}$$

Therefore $h_0, h_1 - h_0, h_2 - h_1, \dots, h_{Ld/2} - h_{Ld/2-1}$
is an 0-sequence.

Thm (Stanley): If Δ is the boundary of a simplicial polytope, then $k[\Delta]/\langle \theta_1, \dots, \theta_d \rangle$ has the hard Lefschetz property.

sketch

Proof: Consider a toric variety whose fan is Δ . Then $k[\Delta]/\langle \theta_1, \dots, \theta_d \rangle$ is isomorphic to the cohomology of this variety. There is a Kähler structure on the toric variety, which implies Hard Lefschetz for $k[\Delta]/\langle \theta_1, \dots, \theta_d \rangle$.

Liter combinational proof by McMullen, using
the polytope algebra.

Thm (g-conjecture): A sequence $h_0, \dots, h_d$ is the h -vector of the boundary of a simplicial polytope iff

- $h_k = h_{d-k}$ for all k
- $h_0, h_1 - h_0, \dots, h_{\lfloor d/2 \rfloor} - h_{\lfloor d/2 \rfloor - 1}$
is an O-sequence.

(Sufficiency shown by Billera-Lee)

Thm (Adiprasito-Papadakis-Petrorou): If Δ is a simplicial sphere and k has characteristic 2, then $k[\Delta]/\langle \theta_1, \dots, \theta_d \rangle$ has the hard Lefschetz property.

This implies the g -conjecture for simplicial spheres.